

COMMAND NEXUS · CIT 140 · WEEK 1

Problem Brief

Meridian Forge Robotics · The Process Below

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SECTION 01

Visible Failure

MR-17 accepted the TEST_K44_START command at 02:13:07. The repeat cycle began at 02:13:09. When the operator sent a stop request at 02:14:02, it returned ACTIVE_INSTRUCTION_PENDING, so the test did not immediately stop. The incident log shows that the stop request did not prevent the repeat cycle from continuing. Source: Incident Log, Nia Voss.

SECTION 02

Verified Facts

MR-17 accepted the TEST_K44_START command at 02:13:07, and the repeat cycle began at 02:13:09. Source: Incident Log, Nia Voss.

MR-17 has no active test assignment, and Test K-44 does not appear in the current test catalog. Source: Current Scheduling Console.

Test K-44 was marked retired 186 days before the incident and was removed from the active workflow. Source: Maintenance Record, Maintenance Archive.

SECTION 03

Assumptions and Open Questions

Nia reported that MR-17 behaved normally before the incident, but previous incident logs or operating records are needed to verify this. Source: Nia Voss, QA Lab.

It is an open question whether Test K-44 was removed from every service and schedule, because the maintenance record does not confirm that every copy, service, or schedule was removed. Source: Maintenance Record, Maintenance Archive.

SECTION 04**System Environment Map****SECTION 04A****People and Roles**

Nia Voss is the Night QA Supervisor who reported the incident. Elias Chen is a Robotics Systems Engineer who provided information about the command and possible restart. Mara Ilyan is the Production Operations Manager who confirmed that production has paused in the affected QA cell. The QA and production teams are affected by the incident.

SECTION 04B**Processes and Rules**

MR-17 should follow its current test assignment and scheduling process. Test K-44 was retired and removed from the active workflow, but the current records do not show an active assignment for MR-17. An operator also sent a stop request after the test began.

SECTION 04C**Information and Data**

The important information includes the incident log, current scheduling console, maintenance record, command telemetry, stop-request status, and records showing when TEST_K44_START was accepted and the repeat cycle began.

SECTION 04D**Technology and Tools**

MR-17 is the affected robot. The command telemetry identifies the source label as legacy_sync.exe and shows that the command was validly formatted. The current scheduling console and test catalog are also relevant system tools. Source: Command Telemetry, Elias Chen.

SECTION 04E

Environment and Constraints

Production has been paused in the affected QA cell. The investigation should avoid assuming a restart is safe or treating an unverified cause as established. The available records do not yet show whether K-44 was removed from every service or schedule.

SECTION 05

Investigation Boundary

Inside the Investigation

The team will examine the command and system logs, the source of the TEST_K44_START command, and whether K-44 is still present in another service or schedule. The team should also compare the incident records with the current scheduling and maintenance records to determine how the retired test was triggered.

Outside for Now

The cause of the incident and a restart should remain outside the investigation for now because the available evidence does not prove what caused the incident or that a restart would safely resolve it. The team should gather more evidence before choosing a solution.

SECTION 06

Organizational Impact

According to Mara, production has been paused in the affected QA cell. This means work in that area is temporarily stopped while the incident is being investigated. Source: Mara Ilyan, Production Operations Manager.

SECTION 07

Problem Statement

MR-17 is running the retired K-44 test even though the current scheduling console shows no active test assignment and K-44 is not in the current test catalog. The stop request did not immediately stop the repeat cycle, and production in the affected QA cell has been paused. The available evidence does not yet establish how the retired test was triggered.

SECTION 08

Next Investigation Step

I would review the command telemetry and system logs to determine which service or process sent the TEST_K44_START command and whether K-44 is still referenced by another service or schedule. This could identify how the retired test was triggered. Source: Command Telemetry, Elias Chen, Robotics Systems Engineer.

SHARED ARTIFACT

Working Document Link

<https://1drv.ms/f/c/afa0d5ff29b575eb/IgC6Ya8ZqFUIR7KYxCSidvunAXBTRrPE1viJlqNHO0vLt7s?e=Xahrce>

WEEK 1 STOPPING RULE

Define before designing.

This Problem Brief records the problem that requires investigation. It does not recommend a solution.

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